THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 9 & 10: **Life Processes in Plants, Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** carry out photosynthesis	site of food making
2	**(b)** carbon dioxide and water	plus sunlight
3	**(d)** warm alcohol	decolourises leaf
4	**(c)** root hairs	absorbing hairs
5	**(c)** energy	uses oxygen
6	**(b)** reflects sunlight	non-luminous
7	**(d)** wooden board	blocks all light
8	**(c)** larger	more light blocked
9	**(c)** same	equal size
10	**(c)** inverted	light crosses at pinhole

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Plants that take food from a host (1); Cuscuta (1).	1+1
12	By trapping and digesting insects (2).	2
13	A box with a tiny hole forming an inverted image on a screen (2).	2
14	Rays from top and bottom cross at the hole (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No chlorophyll, climbs on host (1.5); absorbs host's food (1.5).	1½+1½
16	Light does not need a medium (3).	3
17	Make own food (1.5); food source for all consumers (1.5).	1½+1½
	OR: Pots A and B in sunlight, one watered (1.5); watered grows better (1.5).	1½+1½
18	Look through each material at an object (1.5); classify by clarity (1.5).	1½+1½
	OR: Mirror reflects the torch beam to other areas (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Root nodules (2). (ii) Fixes nitrogen (2). (iii) Crop rotation (1).	2+2+1
	OR: (i) For nitrogen (1). (ii) Slippery pitcher (1). (iii) Yes (1). (iv) Insectivorous (2).	1+1+1+2
20	(i) Dark patch vs picture (2). (ii) Erect virtual vs inverted real (2). (iii) Pinhole image (1).	2+2+1
	OR: (i) Gaps act as pinholes (1). (ii) Pinhole camera (1). (iii) Yes, bigger gap bigger spot (1). (iv) Straight-line travel (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 9 & 10: **Life Processes in Plants, Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** dead and decaying matter	no chlorophyll
2	**(c)** green and non-green patches	chlorophyll in patches
3	**(a)** chemical energy	stored in glucose
4	**(a)** transpiration	pull from leaves
5	**(d)** oxygen	just like animals
6	**(d)** 45° to the tube	parallel to each other
7	**(d)** vacuum	about 3×10^8 m/s
8	**(d)** many rays	bundle of rays
9	**(a)** 1 m/s (so closing speed is 2 m/s)	equal speeds
10	**(d)** splitting of white light	dispersion

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Organisms that feed on dead matter (1); mushroom (1).	1+1
12	Fixes atmospheric nitrogen in root nodules (2).	2
13	To see objects over obstacles using two mirrors (2).	2
14	Multiple reflections in angled mirrors (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Day: photosynthesis (1); night: respiration only (1); transpiration more by day (1).	1+1+1
16	Laterally inverted (1.5); because of left-right reversal (1.5).	1½+1½
17	Removes green colour so iodine colour is visible (3).	3
	OR: CO_2 enters and O_2 leaves through stomata (1.5); reverse in respiration (1.5).	1½+1½
18	Direct a narrow beam on a mirror (1.5); measure angles of incidence and reflection (1.5).	1½+1½
	OR: Dressing, vehicles (1.5); periscopes, torches (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Blade, veins, petiole (2). (ii) Transport (2). (iii) Large area (1).	2+2+1
	OR: (i) Photosynthesis (2). (ii) Stored as starch (1). (iii) Iodine (2).	2+1+2
20	(i) Long to short (2). (ii) Sun's height changes (2). (iii) Time from shadow (1).	2+2+1
	OR: (i) 4 m (2). (ii) 2 m (2). (iii) Image is as far behind as object is in front (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 9 & 10: **Life Processes in Plants, Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** stomata	on leaf surface
2	**(c)** glucose	later stored as starch
3	**(b)** sunlight	energy source
4	**(d)** transpiration	through stomata
5	**(b)** Cuscuta (amarbel)	depends on host
6	**(b)** straight lines	rectilinear propagation
7	**(b)** an opaque object	needs source, object, screen
8	**(d)** far away	less spread
9	**(c)** the distance of the object from the mirror	behind mirror
10	**(d)** screen	translucent screen

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To release energy from food for growth (2).	2
12	Large surface to receive light and exchange gases (2).	2
13	It reflects sunlight (2).	2
14	Sun is higher in the sky at noon (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Plants respire at night using oxygen (3).	3
16	Smooth water reflects regularly (1.5); ripples scatter light (1.5).	1½+1½
17	Make food by photosynthesis (3).	3
	OR: No gas exchange or transpiration (1.5); photosynthesis stops (1.5).	1½+1½
18	All light blocked (3).	3
	OR: Many images (1.5); repeated reflections (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Pale, weak (1). (ii) No photosynthesis (1). (iii) Yes (1). (iv) Place in sunlight (2).	1+1+1+2
	OR: (i) Autotrophs, parasites, insectivorous, saprotrophs (2). (ii) Mango, Cuscuta, pitcher, mushroom (2). (iii) Decompose waste (1).	2+2+1
20	(i) Mirror (1). (ii) Smooth surface (1). (iii) Regular and diffused (1). (iv) Still water, rough road (2).	1+1+1+2
	OR: (i) Left-right reversal (2). (ii) Readable in rear-view mirror (2). (iii) Fire engine label (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET D

Chapters 9 & 10: **Life Processes in Plants, Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** legumes (pulses)	fix nitrogen
2	**(a)** no blue-black colour	no chlorophyll
3	**(a)** chloroplasts	contain chlorophyll
4	**(d)** make its own food	depends on others
5	**(c)** sunlight	needed for food making
6	**(c)** multiple reflections	mirror strips
7	**(b)** regular reflection	clear image
8	**(d)** is large and dark	near source
9	**(a)** many images	repeated reflection
10	**(d)** almost no shadow	light passes through

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Absorb water and minerals from soil (2).	2
12	Cannot photosynthesise and make food (2).	2
13	Regular from smooth surface gives clear image (1); diffused from rough surface scatters light (1).	1+1
14	Image becomes blurred and less sharp (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Petiole holds the leaf (1.5); veins carry water and food (1.5).	1½+1½
16	Light travels in straight lines (3).	3
17	Make food from simple substances (3).	3
	OR: Chlorophyll (1.5); larger absorbing surface (1.5).	1½+1½
18	Orientation changes the shadow shape (1.5); e.g. coin edge-on looks line (1.5).	1½+1½
	OR: Allows vision (1.5); needed for photosynthesis (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Root hairs (2). (ii) Large surface (2). (iii) Growth (1).	2+2+1
	OR: (i) Only producers (2). (ii) Adds O_2, removes CO_2 (2). (iii) Balance gases (1).	2+2+1
20	(i) Straight, reflected, forms shadows (2). (ii) Card-hole and mirror activity (2). (iii) Vision, energy (1).	2+2+1
	OR: (i) Travels in vacuum (2). (ii) Through space (2). (iii) Sound needs a medium (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.